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Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

AS FURTHER MATHEMATICS

Paper 2 Mechanics

Friday 19 May 2023

Afternoon

Time allowed: 1 hour 30 minutes

Materials

- You must have the AQA Formulae and statistical tables booklet for A-level Mathematics and A-level Further Mathematics.
- You should have a graphical or scientific calculator that meets the requirements of the specification.
- You must ensure you have the other optional Question Paper/Answer Book for which you are entered (**either** Discrete **or** Statistics). You will have 1 hour 30 minutes to complete **both** papers.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer each question in the space provided for that question. If you require extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do **not** write outside the box around each page or on blank pages.
- Show all necessary working; otherwise marks for method may be lost.
- Do all rough work in this book. Cross through any work that you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 40.

Advice

- Unless stated otherwise, you may quote formulae, without proof, from the booklet.
- You do not necessarily need to use all the space provided.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
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6	
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9	
TOTAL	



J U N 2 3 7 3 6 6 2 M 0 1

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Answer **all** questions in the spaces provided.

Do not write
outside the
box

- 1** A particle moves along the x -axis under the action of a force, F newtons, where

$$F = 3x^2 + 5$$

Find the work done by the force as the particle moves from $x = 0$ metres to $x = 2$ metres.

Circle your answer.

[1 mark]

12 J

17 J

18 J

34 J

- 2** Two particles are moving directly towards each other when they collide.

Given that the collision is perfectly elastic, state the value of the coefficient of restitution.

Circle your answer.

[1 mark]

$e = -1$

$e = 0$

$e = \frac{1}{2}$

$e = 1$

- 3** A stone of mass 0.2 kg is thrown vertically upwards with a speed of 10 m s^{-1}

Find the initial kinetic energy of the stone.

Circle your answer.

[1 mark]

1 J

5 J

10 J

20 J



4 Reena is skating on an ice rink, which has a horizontal surface.

She follows a circular path of radius 5 metres and centre O

She completes 10 full revolutions in 1 minute, moving with a constant angular speed of ω radians per second.

The mass of Reena is 40 kg

4 (a) Find the value of ω

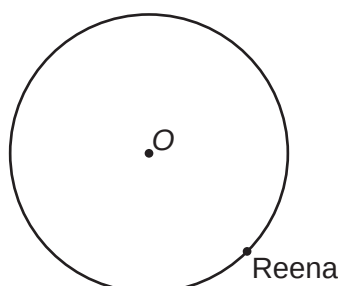
[1 mark]

4 (b) (i) Find the magnitude of the horizontal resultant force acting on Reena.

[2 marks]

4 (b) (ii) Show the direction of this horizontal resultant force on the diagram below.

[1 mark]



Turn over ►



- 5** An impulse of $\begin{bmatrix} -5 \\ 12 \end{bmatrix} \text{ N s}$ is applied to a particle of mass 5 kg which is moving with velocity $\begin{bmatrix} 6 \\ 2 \end{bmatrix} \text{ m s}^{-1}$

- 5 (a)** Calculate the magnitude of the impulse.

[1 mark]

- 5 (b)** Find the **speed** of the particle immediately after the impulse is applied.

[3 marks]



- 6** A ball is thrown with speed u at an angle of 45° to the horizontal from a point O

When the horizontal displacement of the ball is x , the vertical displacement of the ball above O is y where

$$y = x - \frac{kx^2}{u^2}$$

- 6 (a)** Use dimensional analysis to find the dimensions of k

[3 marks]

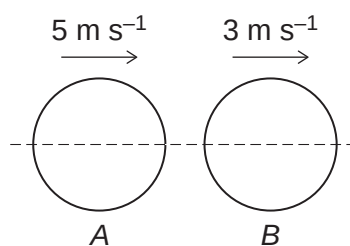
- 6 (b)** State what can be deduced about k from the dimensions that you found in part (a).

[1 mark]

Turn over ►



- 7** Two smooth, equally sized spheres, A and B , are moving in the same direction along a straight line on a smooth horizontal surface, as shown in the diagram below.



The spheres subsequently collide.

Immediately after the collision, A has speed 2.5 m s^{-1} and B has speed 3.5 m s^{-1}

The coefficient of restitution between the spheres is e

- 7 (a) (i)** Show that A does not change its direction of motion as a result of the collision.

[2 marks]

- 7 (a) (ii)** Find the value of e

[1 mark]



7 (b) Given that the mass of B is 0.6 kg, find the mass of A

[3 marks]

Turn over for the next question

Turn over ►



- 8 (b)** If Omar is not modelled as a particle, explain the effect of revising this assumption on your answer to part **(a)**.

[2 marks]

Turn over for the next question

Turn over ►



- 9** Christina is investigating two possible models for the resistance force, R newtons, acting on a car when it is moving in a straight line on a horizontal road.

She uses the following information in both models:

- Mass of the car is 1000 kg
- Maximum power of the engine is 51 kW
- With maximum power, the car accelerates at 4.9 m s^{-2} when travelling at 10 m s^{-1}

- 9 (a)** For Model 1, Christina assumes that R is constant.

- 9 (a) (i)** Show that $R = 200$

[3 marks]

- 9 (a) (ii)** Hence, calculate the maximum possible speed of the car using Model 1.

[2 marks]



9 (b) For Model 2, Christina assumes that $R = kv$, where k is a constant and $v \text{ m s}^{-1}$ is the speed of the car.

9 (b) (i) Find the value of k

[2 marks]

9 (b) (ii) Hence, calculate the maximum possible speed of the car using Model 2.

[2 marks]

9 (c) Evaluate each model, giving reasons for your answers.

[2 marks]

Model 1 _____

Model 2 _____

9 (d) Suggest a further model for R that Christina could consider.

[1 mark]

END OF QUESTIONS



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